WORKED SOLUTION

STEP III 2011, Question 10

Mechanics • Elastic Strings and SHM

STEP III 2011, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q10

The key idea

No external horizontal force acts on the pair, so their centre of mass moves at a constant speed fixed by the impulse. Relative to the centre of mass, the extension of the string obeys simple harmonic motion, and the string is taut for exactly half an oscillation. Splitting the motion into "centre of mass" plus "extension" makes every answer drop out with almost no work.

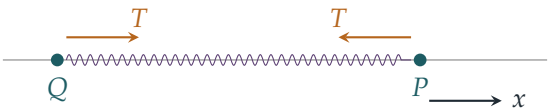
Solution

Model and set-up

The particles move on a smooth horizontal plane, so the only horizontal forces on them are the tensions in the light string, and these are equal and opposite. Take the line QP as the x-axis, pointing from Q to P, with Q starting at $x = 0$ and P at $x = a$. Let x_P and x_Q be their positions at time t after the impulse, and let

$$e = (x_P - x_Q) - a$$

be the extension of the string. Straight after the impulse, $\dot{x}_P = u$, $\dot{x}_Q = 0$ and $e = 0$, with $\dot{e} = u > 0$, so the string at once becomes taut.



While the string is taut, Hooke's law gives tension

$$T = \frac{\lambda e}{a} = \frac{\frac{1}{2}ma\omega^2 e}{a} = \tfrac{1}{2}m\omega^2 e.$$

The tension pulls P back towards Q and pulls Q forward towards P:

$$m\ddot{x}_P = -\tfrac{1}{2}m\omega^2 e, \qquad m\ddot{x}_Q = +\tfrac{1}{2}m\omega^2 e.$$

The centre of mass

Adding the two equations, $\ddot{x}_P + \ddot{x}_Q = 0$. So the centre of mass G, at $\bar{x} = \frac{1}{2}(x_P + x_Q)$, moves with constant velocity equal to its initial value,

$$\dot{\bar{x}} = \tfrac{1}{2}(u + 0) = \tfrac{1}{2}u.$$

(This is just conservation of momentum: the total momentum mu is shared by mass $2m$.)

The extension

Subtracting the equations,

$$\ddot{e} = \ddot{x}_P - \ddot{x}_Q = -\omega^2 e.$$

This is simple harmonic motion with angular frequency ω. With $e(0) = 0$ and $\dot{e}(0) = u$,

$$e = \frac{u}{\omega}\sin\omega t.$$

Ask yourself Why is the angular frequency ω and not $\omega/\sqrt{2}$ or $\sqrt{2}\,\omega$?

Each particle feels the force $\frac{1}{2}m\omega^2 e$, but e changes because *both* particles move: their accelerations add when we subtract the equations. Equivalently, the reduced mass of two equal masses m is $\frac{1}{2}m$, and the string's stiffness is $\lambda/a = \frac{1}{2}m\omega^2$, so $\omega^2 = \frac{1}{2}m\omega^2 / \frac{1}{2}m$. The modulus was chosen to make this come out neatly.

The string is next of length a when $e = 0$ again, that is when $\sin\omega t = 0$ for the first time with $t > 0$:

$$t_1 = \frac{\pi}{\omega}.$$

Throughout $0 < t < \pi/\omega$ we have $e > 0$, so the string really is taut and the equations above hold.

Distance travelled

Writing each position as the centre of mass plus or minus half the separation,

$$x_P = \bar{x} + \tfrac{1}{2}(a + e), \qquad x_Q = \bar{x} - \tfrac{1}{2}(a + e).$$

At t_1, $e = 0$ again, so each particle has moved by exactly the same amount as G, namely

$$\tfrac{1}{2}u \cdot \frac{\pi}{\omega} = \frac{\pi u}{2\omega}.$$

To be sure this displacement is the distance travelled, we check that neither particle ever turns back. Differentiating,

$$\dot{x}_P = \tfrac{1}{2}u + \tfrac{1}{2}\dot{e} = \tfrac{1}{2}u(1 + \cos\omega t), \qquad \dot{x}_Q = \tfrac{1}{2}u - \tfrac{1}{2}\dot{e} = \tfrac{1}{2}u(1 - \cos\omega t),$$

and both are ≥ 0 for all t. So each particle moves only forwards, and each has travelled a distance $\dfrac{\pi u}{2\omega}$, as required.

The speeds

At $t_1 = \pi/\omega$, $\cos \omega t_1 = -1$, so

$$\dot{x}_P = 0, \qquad \dot{x}_Q = u.$$

So P is momentarily at rest and Q moves with speed u towards P. The particles have exactly swapped velocities, just as in a perfectly elastic collision of equal masses. That's no accident: momentum is conserved, and the elastic energy stored has been completely returned as kinetic energy.

The time to collision

After t_1, Q is moving towards P, so the separation is less than a. The string is slack and exerts no force, and neither particle feels any horizontal force. Hence P stays at rest and Q moves at constant speed u, starting a distance a from P. It reaches P after a further time a/u. The total time from the impulse to the collision is

$$\frac{\pi}{\omega} + \frac{a}{u}.$$

Checks

- Momentum: at t_1 the total momentum is $m \cdot 0 + mu = mu$, as at the start. Energy: kinetic energy $\frac{1}{2}mu^2$ at both moments, with the string at its natural length both times.
- Dimensions: u/ω is a length and a/u is a time, as they should be.
- We stepped the equations of motion numerically for $m = a = \omega = u = 1$. The string went slack at $t = 3.14159$ with each particle having moved $1.57080 = \pi/2$; the speeds were then 0 and 1, and the collision came at $t = 4.14159 = \pi + 1$.

Common slips

- Treating P as attached to a fixed point, which gives angular frequency $\omega/\sqrt{2}$ and the wrong time. Both ends of the string move.
- Continuing the simple harmonic motion after t_1. Once the string is shorter than a it is slack, so the motion is uniform.
- Forgetting the final stage altogether: the collision happens a further a/u after the string returns to its natural length.

Beyond STEP

Splitting a two-body problem into motion of the centre of mass and motion of the separation, governed by the reduced mass $\mu = m_1 m_2/(m_1 + m_2)$, is one of the most

useful tricks in first-year mechanics. It turns planetary orbits, vibrating diatomic molecules and colliding particles into one-body problems. Here the elastic string acts as a "soft" collision that lasts a time π/ω but has exactly the same outcome as an instantaneous elastic one. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops the two-body reduction and the reduced mass carefully.

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